Table of Contents

..... 1
1. FIR FILTER PARAMETERS 1
2. DIRECT REALIZATION 1
3. CASCADE REALIZATION 1
4. SYMMETRY AND LINEAR PHASE 1
5. FREQUENCY RESPONSE 2
6. NUMERICAL COMPARISON 2
7. DISPLAY RESULTS 2
8. OUTPUT PARAMETER TABLE 3
9. GRAPH 1: IMPULSE RESPONSE 4
10. GRAPH 2: MAGNITUDE RESPONSE 5
11. GRAPH 3: PHASE RESPONSE 6
12. GRAPH 4: GROUP DELAY 7
13. GRAPH 5: NUMERICAL ERROR 8
14. FINAL RESULT 9


```
clc;
clear;
close all;
```

1. FIR FILTER PARAMETERS

```
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];
n = 0:N-1;
```

2. DIRECT REALIZATION

```
yd = filter(b,1,x);
```

3. CASCADE REALIZATION

```
z = roots(b);
yc = b(1)*x;

for k = 1:length(z)
    yc = filter([1 -z(k)],1,yc);
end

yc = real(yc);
```

4. SYMMETRY AND LINEAR PHASE

```
symErr = max(abs(b-fliplr(b)));
D = (length(b)-1)/2;
```

```
if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end
```

5. FREQUENCY RESPONSE

```
[H,w] = freqz(b,1,1024);
phase = unwrap(angle(H));
gd = grpdelay(b,1,1024);
```

6. NUMERICAL COMPARISON

```
err = max(abs(yd-yc));
dev = err/max(abs(yd))*100;
```

7. DISPLAY RESULTS

```
fprintf('\n====================================\n');
fprintf(' FIR FILTER RESULTS\n');
fprintf('====================================\n');

fprintf('Sampling Frequency = %d Hz\n',Fs);
fprintf('Number of Samples = %d\n',N);
fprintf('Coefficient Symmetry = %s\n',symmetry);
fprintf('Symmetry Error = %.3e\n',symErr);
fprintf('Group Delay = %.2f samples\n',D);
fprintf('Maximum Error = %.3e\n',err);
fprintf('Percentage Deviation = %.3e %%\n',dev);

fprintf('\nFIR Coefficients:\n');
disp(b);

fprintf('\nFirst 10 Impulse Response Values\n');
fprintf(' n       Direct       Cascade\n');

for k = 1:10
    fprintf('%2d     %10.6f   %10.6f\n', ...
        n(k),yd(k),yc(k));
end

fprintf('\nCascade Zeros:\n');
disp(z);
```

```
====================================
 FIR FILTER RESULTS
====================================
Sampling Frequency = 1000 Hz
Number of Samples = 128
```

```
Coefficient Symmetry = Symmetric
Symmetry Error = 0.000e+00
Group Delay = 2.00 samples
Maximum Error = 8.327e-16
Percentage Deviation = 2.220e-13 %

FIR Coefficients:
    0.0625    0.2500    0.3750    0.2500    0.0625


First 10 Impulse Response Values
 n       Direct       Cascade
 0       0.062500     0.062500
 1       0.250000     0.250000
 2       0.375000     0.375000
 3       0.250000     0.250000
 4       0.062500     0.062500
 5       0.000000     0.000000
 6       0.000000     0.000000
 7       0.000000     0.000000
 8       0.000000     0.000000
 9       0.000000     0.000000

Cascade Zeros:
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

8. OUTPUT PARAMETER TABLE

```
Parameter = {
    'Direct realization output'
    'Cascade realization output'
    'Impulse response'
    'Phase response'
    'Comparison of realizations'
};

Observation = {
    'Direct FIR filtering'
    'Cascade of first-order sections'
    'FIR coefficient sequence'
    'Linear phase'
    'Numerical equivalence'
};

Value = {
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yc(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('Group delay = %.2f samples',D)
```

```
    sprintf('Maximum error = %.3e',err)
};

Unit_Note = {
    'Amplitude'
    'Amplitude'
    'h[n]'
    'Samples'
    'Numerical precision'
};

Output_Table = table(Parameter,Observation,Value,Unit_Note);

fprintf('\nOUTPUT PARAMETER TABLE\n');
disp(Output_Table);
```

```
OUTPUT PARAMETER TABLE
              Parameter                                  Value
Observation                                                                  Unit_Note
    ____________________________________    ____________________________________
_____________________________________________    ____________________________

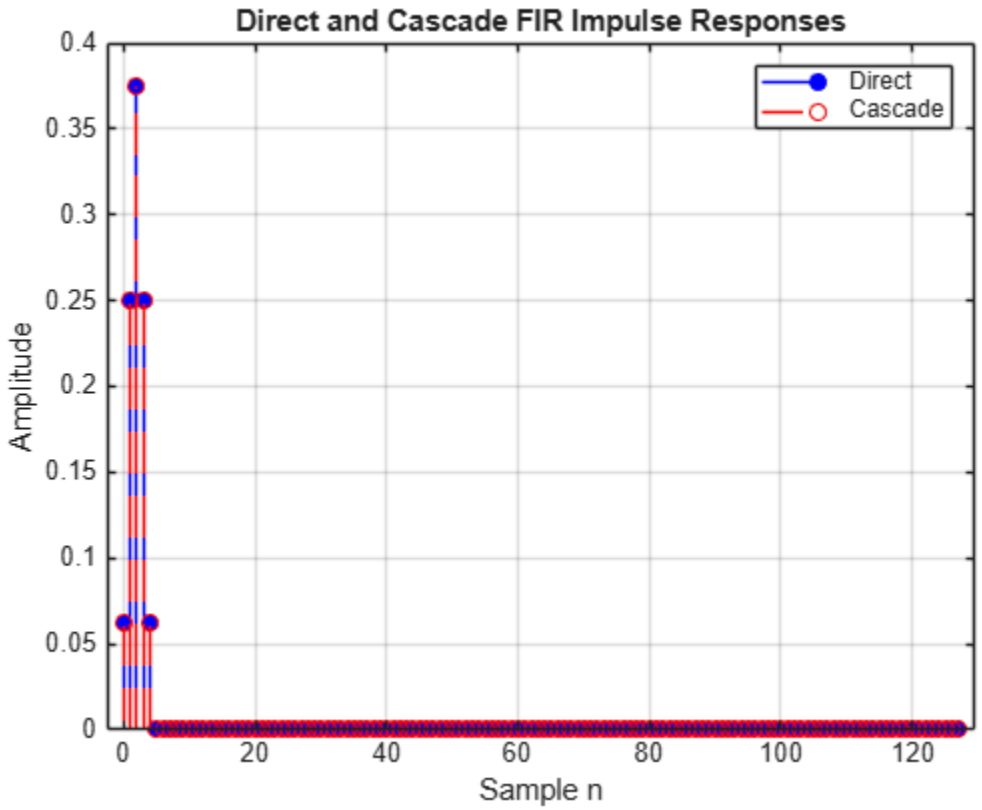
    {'Direct realization output' }    {'Direct FIR filtering'           }
{'[0.0625 0.2500 0.3750 0.2500 0.0625]'}    {'Amplitude'          }
    {'Cascade realization output'}    {'Cascade of first-order sections'}
{'[0.0625 0.2500 0.3750 0.2500 0.0625]'}    {'Amplitude'          }
    {'Impulse response'          }    {'FIR coefficient sequence'       }
{'[0.0625 0.2500 0.3750 0.2500 0.0625]'}    {'h[n]'               }
    {'Phase response'            }    {'Linear phase'                   }
{'Group delay = 2.00 samples'          }    {'Samples'            }
    {'Comparison of realizations'}    {'Numerical equivalence'          }
{'Maximum error = 8.327e-16'           }    {'Numerical precision'}
```

9. GRAPH 1: IMPULSE RESPONSE

```
figure('Name','Impulse Response');

stem(n,yd,'b','filled');
hold on;
stem(n,yc,'r--');

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Direct and Cascade FIR Impulse Responses');
legend('Direct','Cascade');
```

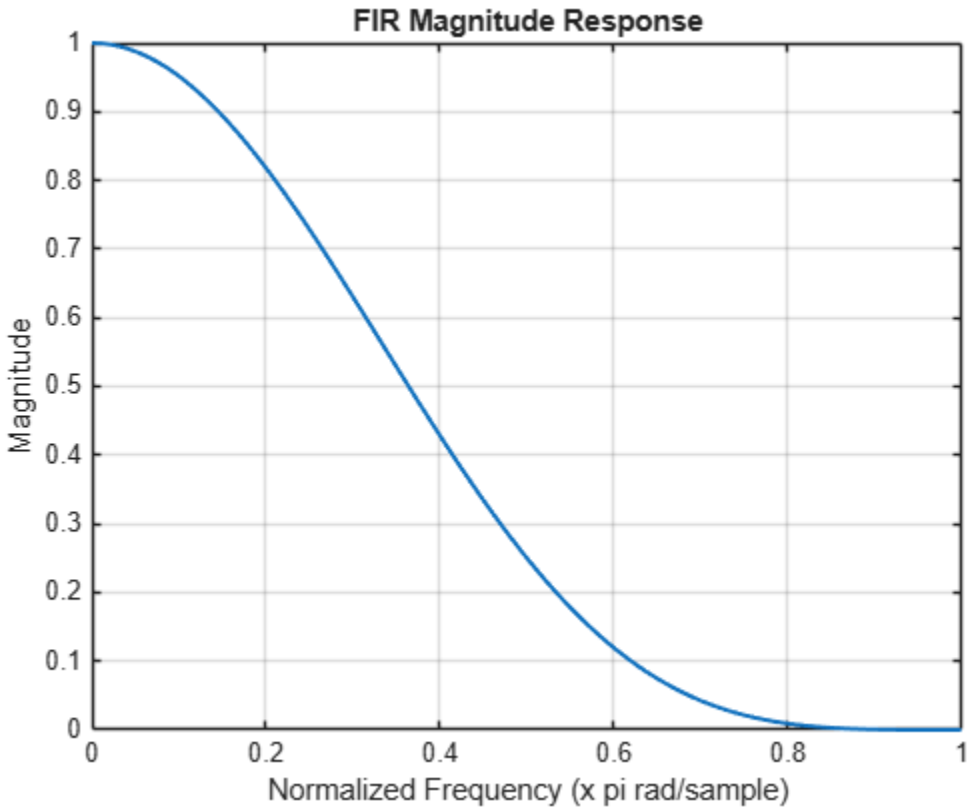


10. GRAPH 2: MAGNITUDE RESPONSE

```
figure('Name','Magnitude Response');

plot(w/pi,abs(H),'LineWidth',1.5);

grid on;
xlabel('Normalized Frequency (x pi rad/sample)');
ylabel('Magnitude');
title('FIR Magnitude Response');
```

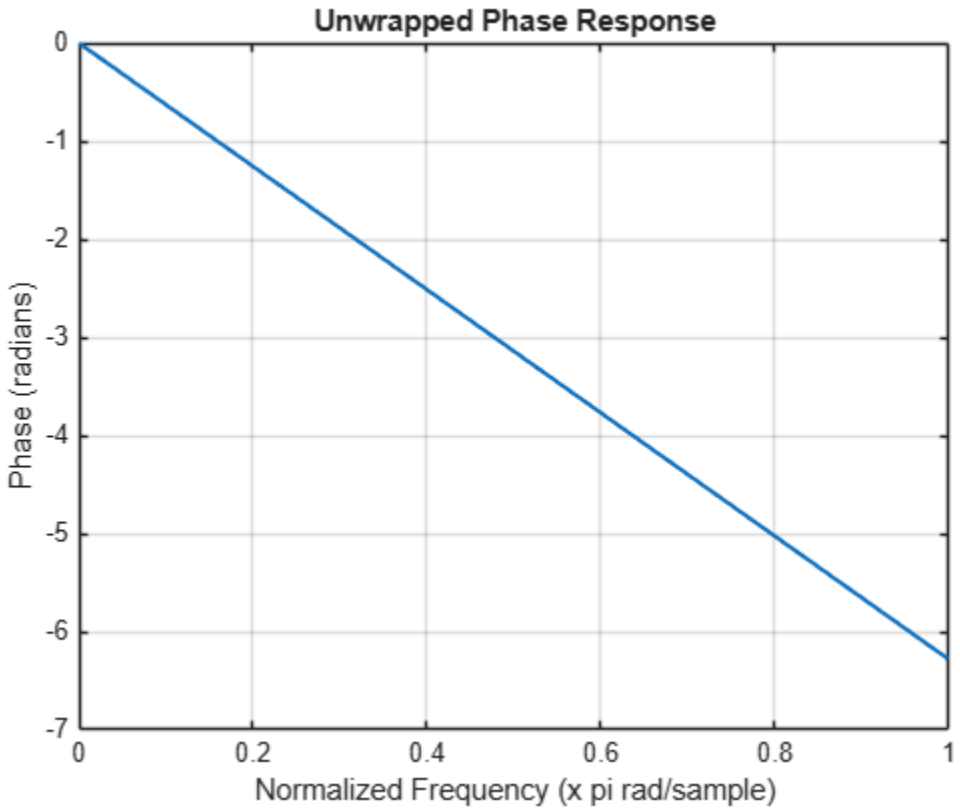


11. GRAPH 3: PHASE RESPONSE

```
figure('Name','Phase Response');

plot(w/pi,phase,'LineWidth',1.5);

grid on;
xlabel('Normalized Frequency (x pi rad/sample)');
ylabel('Phase (radians)');
title('Unwrapped Phase Response');
```

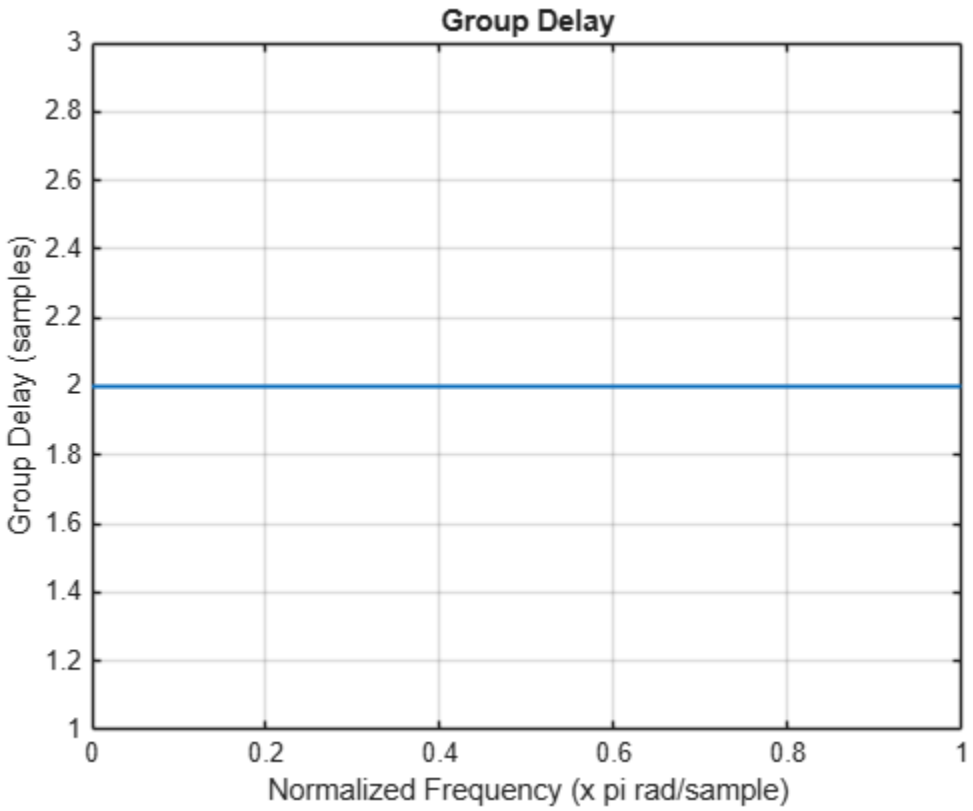


12. GRAPH 4: GROUP DELAY

```
figure('Name','Group Delay');

plot(w/pi,gd,'LineWidth',1.5);

grid on;
xlabel('Normalized Frequency (x pi rad/sample)');
ylabel('Group Delay (samples)');
title('Group Delay');
```



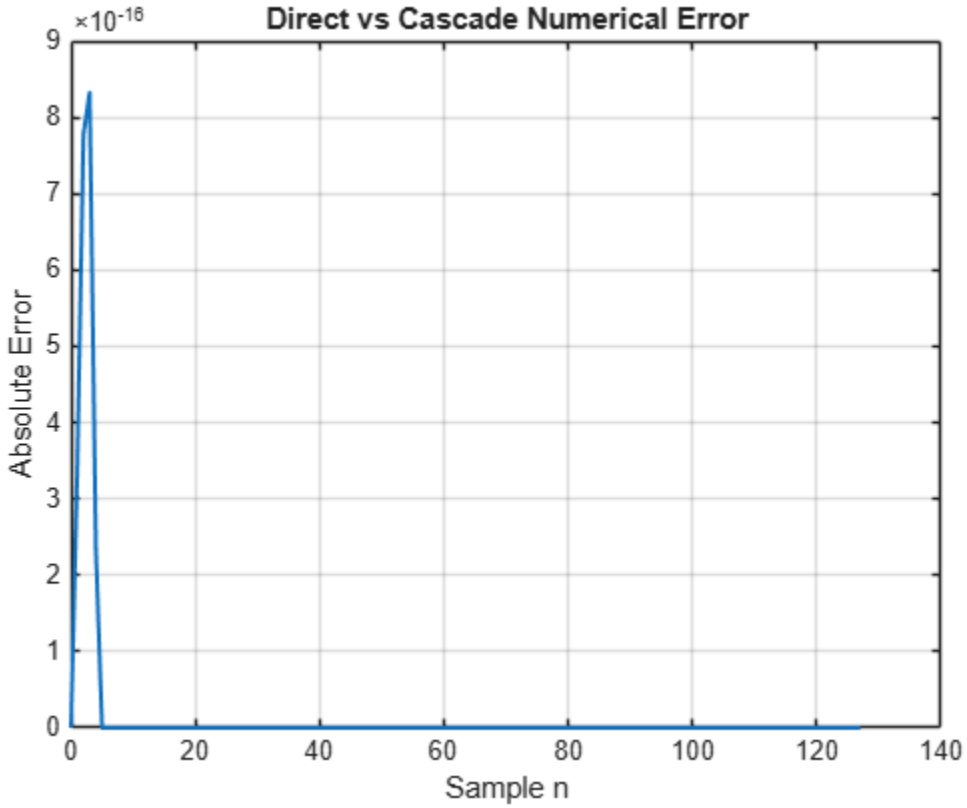
13. GRAPH 5: NUMERICAL ERROR

```
figure('Name','Numerical Error');

plot(n,abs(yd-yc),'LineWidth',1.5);

grid on;
xlabel('Sample n');
ylabel('Absolute Error');
title('Direct vs Cascade Numerical Error');

drawnow;
```



14. FINAL RESULT

```
fprintf('\n====================================\n');
fprintf(' FINAL RESULT\n');
fprintf('====================================\n');

fprintf('1. Direct FIR realization completed.\n');
fprintf('2. Cascade FIR realization completed.\n');
fprintf('3. Coefficients are %s.\n',symmetry);
fprintf('4. Group delay = %.2f samples.\n',D);
fprintf('5. Maximum error = %.3e.\n',err);
fprintf('6. Percentage deviation = %.3e %%.\n',dev);
fprintf('7. Both realizations are numerically equivalent.\n');
```

```
====================================
 FINAL RESULT
====================================
1. Direct FIR realization completed.
2. Cascade FIR realization completed.
3. Coefficients are Symmetric.
4. Group delay = 2.00 samples.
5. Maximum error = 8.327e-16.
6. Percentage deviation = 2.220e-13 %.
7. Both realizations are numerically equivalent.
```

Published with MATLAB® R2026b